DO NOW — RECALL

Data structure selection challenge

Name: _____ Class: _____ Date: _____

1. What is an accumulator?

2. What is a collision?

3. What is a head pointer?

4. Which structure is used for Undo and why?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A variable storing a running total.

2. Model answer

Two keys map to the same initial bucket.

3. Model answer

A reference to the first node.

4. Model answer

A stack: the most recent operation must be undone first.

Vocabulary — know these terms

LIFO — Last In, First Out.

FIFO — First In, First Out.

lookup — Retrieve a value using a key.

balanced BST — Tree kept at relatively small height.

trade-off — Benefit balanced against a drawback.

NEW LEARNING

Read, annotate and explain

Identify the operation

The print job that arrived first must leave first.

Match the structure

Enqueue adds jobs; dequeue removes the oldest one. This is a queue.

Justify and qualify

A stack gives LIFO, which is the wrong order. Other systems may also need priority queues if priority overrides arrival order.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A browser Back button must return to the most recently visited page. Choose a structure.

Predict

The last visited page is the first one that must be removed.

Explain

Use a stack: push visits and pop the most recent page.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Select a structure for processing jobs in arrival order.

Question 2

Choose stack or hash table to retrieve customer records by unique ID. Justify.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare a balanced BST and hash table for lookup and sorted output.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hash table is always O(1) even in the worst case.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.